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Research Article

Maternal vaginal colonization screening for term singleton pregnancy: comparative evaluation of metagenomic next-generation sequencing (mNGS) versus real-time quantitative PCR (qPCR)

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ABSTRACT

Objective: To investigate the distribution characteristics of potential high-risk pathogens for early-onset neonatal infection in maternal vaginal secretions, and to perform a head-to-head comparative evaluation of detection performance for target pathogens between metagenomic next-generation sequencing (mNGS) and real-time quantitative polymerase chain reaction (qPCR), with conventional bacterial culture as the reference standard.

Methods: A total of 294 valid maternal vaginal secretion samples were prospectively collected and tested in parallel using qPCR, mNGS, and conventional bacterial culture. The Chi-square test was used to compare the differences in pathogen detection rates among the three methods. Receiver operating characteristic (ROC) curve was plotted to calculate the area under the curve (AUC) and 95% confidence interval (CI), to systematically evaluate the detection performance of the two methods for target pathogens.

Results: The spectrum of potential early-onset neonatal pathogens in maternal vaginal secretions, ranked by detection rate, was as follows: *Staphylococcus aureus*, *Streptococcus agalactiae*, *Ureaplasma urealyticum*, *Listeria monocytogenes*, and *Campylobacter fetus*. The detection rates of these target pathogens by qPCR, mNGS, and bacterial culture showed high consistency, with no statistically significant difference in detection rates among the three methods (all $P > 0.05$). ROC curve analysis showed that the AUC values of both qPCR and mNGS for the above major pathogens were all above 0.90, which were significantly different from the null hypothesis of $AUC = 0.5$ (all $P < 0.05$), indicating good detection performance; while there was no significant difference in AUC values between qPCR and mNGS (all $P > 0.05$). In addition, *Listeria monocytogenes* (3 cases) and *Campylobacter fetus* (1 case) were only detected by qPCR and mNGS, while not isolated by conventional culture.

Conclusion: This head-to-head comparative study confirms that both mNGS and targeted qPCR have high accuracy and consistency for detecting potential early-onset neonatal pathogens in maternal vaginal secretions. We propose a tiered antenatal screening strategy for maternal vaginal pathogenic colonization: qPCR is recommended as the first-line tool for routine antenatal screening due to its high cost-effectiveness and rapid turnaround time, while mNGS is reserved for

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high-risk pregnant women (e.g., preterm premature rupture of membranes, clinical chorioamnionitis), culture-negative suspected infection cases, or scenarios requiring comprehensive pathogen profiling, to take full advantage of its unbiased, broad-spectrum detection capability. This integrated screening strategy requires further prospective validation with paired neonatal clinical outcome data to confirm its value in the prevention and early intervention of early-onset neonatal infection.

1. Introduction

Perinatal and neonatal infections, which are primarily transmitted through vertical transmission of maternal vaginal microbiota, pose serious threats to infant health [1]. Pathogens can invade neonates via placenta, contact exposure during childbirth, and postpartum interactions [2,3]. Vertical transmission of colonizing pathogens from the maternal vaginal tract may lead to severe complications such as sepsis, pneumonia, and even meningitis. Therefore, systematic characterization of the pathogenic spectrum in maternal vaginal secretions is crucial for early intervention and targeted treatment of neonatal infections.

Current routine microbiological diagnostic methods mainly include bacterial culture, biochemical identification, and serological tests [4,5]. However, traditional techniques have notable limitations: extended turnaround time of culture-based methods is a bottleneck for urgent clinical needs, and their suboptimal sensitivity in detecting fastidious bacteria and polymicrobial infections often results in missed diagnoses. Thus, there is an urgent need to adopt detection technologies characterized by rapidness, accuracy, and comprehensiveness to advance the diagnosis and treatment of neonatal infections.

Metagenomic next-generation sequencing (mNGS), a recently developed unbiased pathogen detection approach, enables comprehensive screening of diverse microorganisms, including bacteria, viruses, fungi, and mycoplasma, in clinical specimens [6]. Leveraging its advantages of speed, high sensitivity, and high throughput, mNGS can be used to not only identify pathogen species and quantify load within shorter timeframes but also to complement information on rare pathogens undetectable via conventional methods [7,8]. Although existing studies demonstrate promising applications of mNGS in pediatric central nervous system infections among other areas, research focusing on the use of mNGS for screening maternal vaginal secretions for common neonatal pathogens remains insufficient and warrants further exploration.

Consequently, this study employs mNGS to investigate the composition of high-risk neonatal pathogens in maternal vaginal secretions and systematically compares its performance with a targeted qPCR approach. The aim is twofold: to evaluate the practical efficacy of mNGS in this specific clinical context and to explore its potential to uncover additional diagnostic value beyond conventional targeted methods, thereby informing a more effective diagnostic pathway.

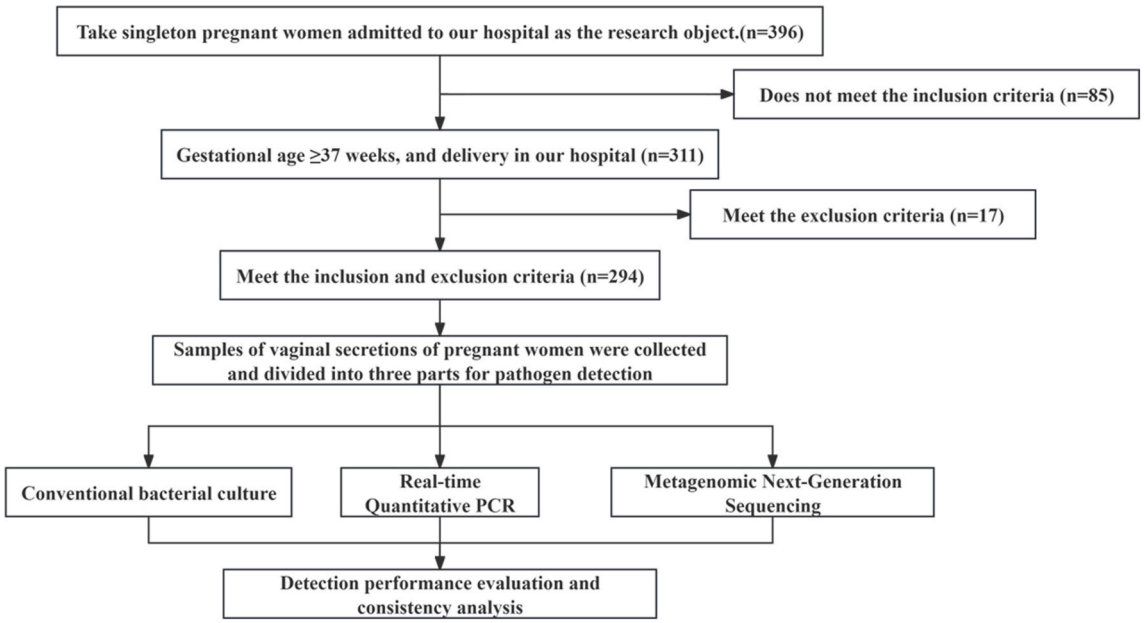


Fig. 1. Study flow chart.

2. Materials and methods

2.1. Study flow chart

2.2. Clinical data

2.2.1. Study subjects

All vaginal secretion samples used in this study were prospectively collected from singleton pregnant women admitted to Dongguan Tungwah Hospital between December 2023 and December 2024, which was consistent with the prospective study design throughout the whole research. A total of 396 singleton pregnant women were initially screened for enrollment. The pre-specified inclusion criteria were: (1) singleton pregnancy; (2) gestational age ≥37 weeks; (3) planned delivery in our hospital; (4) complete clinical data available.85 participants were excluded in the first round as they did not meet the above inclusion criteria, leaving 311 eligible participants for preliminary screening. Subsequently, 17 participants were excluded from the final analysis according to the pre-specified exclusion criteria, including patients with preterm premature rupture of membranes (PPROM), premature rupture of membranes (PROM), clinical chorioamnionitis, or other severe perinatal complications, as these cases had no clinical indication for perinatal pathogenic screening and may interfere with the head-to-head performance comparison among the three detection methods. Finally, a total of 294 valid vaginal secretion samples from eligible pregnant women were included in the final analysis, and all samples were tested in parallel by real-time quantitative polymerase chain reaction (qPCR), metagenomic next-generation sequencing (mNGS), and conventional bacterial culture (see Fig. 1).

2.3. Methods

2.3.1. Sampling method

All vaginal secretion samples were collected for research purposes only rather than as a part of routine clinical care. After urination, pregnant women were placed in the lithotomy position. Following standard vulvar disinfection, a sterile swab was inserted into the posterior vaginal fornix and gently rotated for 15–20 s to ensure adequate absorption of the sample. Subsequently, the swab was immediately placed into a specimen transport tube and delivered to the laboratory within 1 h. All samples were collected at 36–39 weeks of gestation during routine late prenatal check ups, before delivery and without clinical symptoms of infection.

2.3.2. Pathogen detection strategy

(1) Conventional Culture and Identification:

Standard culture media (such as blood agar, chocolate agar, MacConkey agar, etc.) were used for the isolation and identification of target bacterial strains.

Table 1
Primers and detection limits of multiple fluorescent quantitative PCR.

Gene name	Primer name	Primer sequence	Minimum detection limit (fg)	Standard deviation
cfb	F	CGGTTAATGAGGCTATTAC	1.17	0.57
	R	GCTTCAATCACATCTGTTA		
	T	CGCCATCGCCGTCTTCATCA		
dltS	F	CCATGCCATTAATTAGACC	14.33	0.23
	R	TGCAGGACATGAACTAAA		
	T	TCCACTTGCTATCATCTCCGCTAATAAT		
sapB2	F	GCAGGAAATACTCTAGGA	32.23	2.82
	R	GGGCAGTTATCTCTACTAC		
	T	TCTCTGAACCTACATAAGTAACTTCGT		
cadF	F	CCGATACTATCTCTGTTTATG	32.17	7.26
	R	GTTCCAGGACATCTATCTA		
	T	CGCCATCGCCGTCTTCATCA		
nuc	F	GCGTAAATAGAAGTGGTTC	1.29	0.004
	R	GTCTGAATGTCATTGGTTG		
	T	TAACCGTATCACCATCAATCGCTT		
hlyA	F	CCGCAAAAGATGAAGTTC	1.29	0.12
	R	CAGCTAATTCATTGTCTTTTAAG		
	T	CGGTAACCTCGGAGACTTACG		
ure	F	ACATTAGCACCAACATAAGG	3.56	0.29
	R	TGATAAACACCCTGATTT		
	T	GCAACATTTAGTCCAGATTTAGT		

(2) mNGS Detection:

Total DNA was extracted and a library was prepared. Sequencing was performed on the Illumina platform (PE150). Human-derived sequences were filtered out by aligning to the National Center for Biotechnology Information (NCBI) database, and a threshold of no fewer than 10 uniquely mapped reads was set for pathogen determination.

(3) qPCR Experiments:

Specific primers and probes were designed based on high-risk neonatal pathogens identified from the mNGS sequencing data. Targets included genes of *Streptococcus agalactiae* (*cfb*, *dltS*), *Campylobacter fetus* (*sapB2*, *cadF*), *Staphylococcus aureus* (*nuc*), *Listeria monocytogenes* (*hlyA*), and *Ureaplasma urealyticum* (*ure*) (Table 1). The reaction mixture was as follows: 10 μL 2X fluorescent quantification reagent (CSP), 0.8 μL upstream and downstream primer (400 nM), 0.4 μL probe (200 nM), and 2 μL template DNA. The amplification conditions were as follows: predenaturation at 95 °C for 5 min, followed by 40 cycles of 95 °C for 15 s, and 60 °C for 40 s. The lowest detection limit (LOD) of each gene was determined using gradient dilution (10 ng-1 fg). For example, the LOD of the dltS gene was calculated to be 14.33 ± 0.23 fg. A total of 165 common strains (including *Escherichia coli*, *Candida*) were tested to verify primer specificity (100% no cross-reactivity).

2.4. Statistical analysis

Data analysis was performed using SPSS software (version 21.0) and MedCalc software (version 20.0). The Chi-square test was used to compare the differences in pathogen detection rates among qPCR, mNGS and conventional bacterial culture. Receiver operating characteristic (ROC) curves were plotted to calculate the area under the curve (AUC) and 95% confidence interval (CI) of qPCR and mNGS, with conventional bacterial culture as the reference standard. DeLong's test was used to compare the differences in AUC values between the two methods, to evaluate the difference in detection performance between qPCR and mNGS. A two-sided $P < 0.05$ was considered statistically significant, and $P > 0.05$ indicated no statistically significant difference between the compared groups.

3. Results

3.1. Comparison of pathogen detection rates

The detection rates for *Staphylococcus aureus*, *Streptococcus agalactiae*, and *Ureaplasma urealyticum* using qPCR, bacterial culture, and mNGS showed high consistency ($P > 0.05$) (Table 2). The number of positive cases based on the results of qPCR, culture, and mNGS was 114, 121, and 126 respectively, for *S. aureus*; 84, 89, and 91, respectively, for *S. agalactiae*; and 51, 55, and 52, respectively, for *U. urealyticum*. No statistically significant differences were observed among the groups, indicating that molecular detection methods and conventional culture yielded comparable results in screening for common pathogens. In addition, *Listeria monocytogenes* (three cases) and *Campylobacter fetus* (one case) were detected exclusively using qPCR and mNGS, indicating that these molecular methods can supplement the range of pathogens detected by culture.

3.2. Diagnostic performance of qPCR and mNGS

3.2.1. Staphylococcus aureus

The sensitivity of qPCR and mNGS was 100.00% (95% confidence interval [CI]:96.48–100%) and 96.03% (95% CI: 90.85–98.36%), respectively. Compared with culture, the specificity of these methods was 96.11% (95% CI: 92.17–98.18%) and 100.00% (95% CI: 97.67–100%), and accuracy was 97.62% (95% CI: 94.87–98.92%) and 98.30% (95% CI: 95.43–99.45%), respectively, with no statistically significant difference between the methods ($P > 0.05$, Tables 3 and 4). Both methods demonstrated good diagnostic performance, with areas under the receiver operating characteristic curve (ROC AUC) of 0.929 (95% CI: 0.880–0.978) and 0.917 (95% CI: 0.853–0.980), respectively; all AUC values were at least 0.917 ($P < 0.05$, Fig. 2).

3.2.2. Streptococcus agalactiae

The sensitivity of qPCR and mNGS was 100% (95% CI: 95.54–100%) and 92.31% (95% CI: 84.77–96.83%), respectively. Compared with culture, their specificities were 97.62% (95% CI: 94.54–99.21%) and 100% (95% CI: 98.21–100%), with corresponding accuracy

Table 2
Comparison of the detection results of vaginal secretions by three methods.

Pathogenic bacteria	PCR positive number (%)	Culture positive number (%)	mNGS positive number (%)	χ^2/t-value	*P*-value
Streptococcus agalactiae	84	89	91	0.422	0.810
Ureaplasma urealyticum	51	55	52	0.201	0.905
Staphylococcus aureus	114	121	126	1.022	0.600
Listeria monocytogenes	3	undetected	3	-	-
Campylobacter fetus	1	undetected	1	-	-

Table 3
Analysis of the coincidence rate of detection of *Staphylococcus aureus* by qPCR and mNGS diagnostic methods.

Gold standard, (Positive = 121, Negative = 173)	qPCR Detection		mNGS diagnosis	
	Positive	Negative	Positive	Negative
Positive	114	7	121	0
Negative	0	173	5	168
Total	114	180	126	168

Table 4
Diagnostic value of qPCR and mNGS for *Staphylococcus aureus*.

Inspection mode	Sensitivity (%)	Specificity (%)	Accuracy (%)	Positive predictive value	Negative predictive value
qPCR Detection	114/114 (100.00)	173/180 (96.11)	287/294 (97.62)	114/121 (94.21)	173/173 (100.00)
mNGS diagnosis	121/126 (96.03)	168/168 (100.00)	289/294 (98.30)	121/121 (100.00)	168/173 (97.11)
χ^2	2.880	4.840	0.340	5.296	3.247
P	0.090	0.028	0.560	0.021	0.072

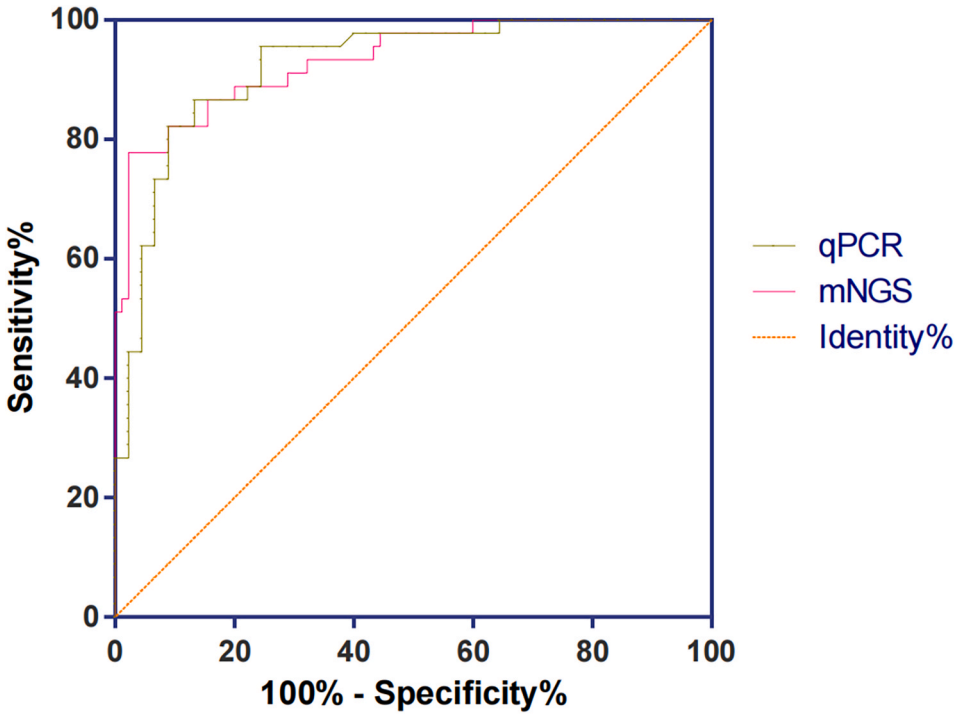


Fig. 2. ROC curve for the diagnosis of *Staphylococcus aureus* by qPCR detection and mNGS diagnosis.

rates of 98.30% (95% CI: 96.29–99.32%) and 97.62% (95% CI: 95.13–98.95%), with no statistically significant difference ($P > 0.05$, Tables 5 and 6). Both methods demonstrated good diagnostic performance, with ROC AUC of 0.929 (95% CI: 0.880–0.978) and 0.917 (95% CI: 0.853–0.980), respectively; all AUC values were at least 0.917 ($P < 0.05$, Fig. 3).

Table 5
Analysis of the coincidence rate of detection of *Streptococcus agalactiae* by qPCR and mNGS diagnostic methods.

Gold standard	qPCR Detection		mNGS diagnosis	
	Positive	Negative	Positive	Negative
Positive	84	5	84	0
Negative	0	205	7	203
Total	84	210	91	203

Table 6
Diagnostic value of qPCR and mNGS for *Streptococcus agalactiae*.

Inspection mode	Sensitivity (%)	Specificity (%)	Accuracy (%)	Positive predictive value	Negative predictive value
PCR Detection	84/84 (100.00)	205/210 (97.62)	289/294 (98.30)	84/89 (94.38)	205/205 (100.00)
mNGS diagnosis	84/91 (92.31)	203/203 (100.00)	287/294 (97.62)	84/84 (100.00)	203/210 (96.67)
χ^2	4.878	3.104	0.085	3.064	5.086
P	0.009	0.078	0.771	0.080	0.024

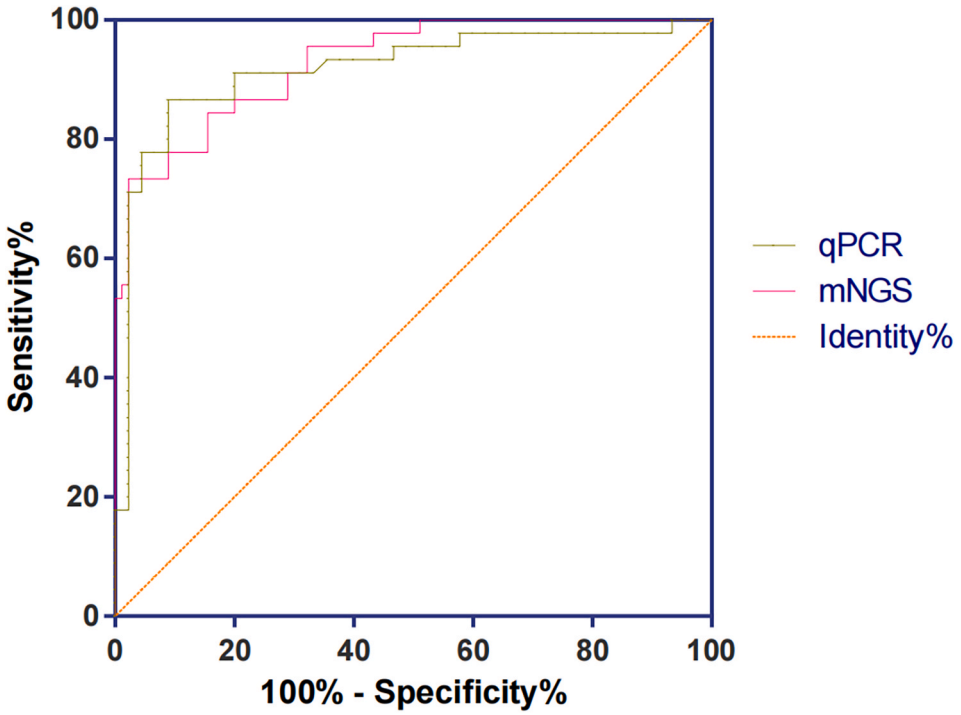


Fig. 3. ROC curve of the diagnosis of *Streptococcus agalactiae* by qPCR and mNGS.

3.2.3. Ureaplasma urealyticum

Both qPCR and mNGS showed a sensitivity of 100% for the detection of *U. urealyticum*. Compared with culture, their specificities were 98.35% (95% CI: 95.32–99.09%) and 98.76% (95% CI: 96.40–99.74%), and their accuracy rates were 98.64% (95% CI: 96.76–99.56%) and 98.98% (95% CI: 97.39–99.70%), respectively, with no statistically significant difference between the two methods ($P > 0.05$, Tables 7 and 8). Both methods demonstrated good diagnostic performance, with ROC AUC of 0.938 (95% CI: 0.889–0.987) and 0.927 (95% CI: 0.867–0.986), respectively; all AUC values were at least 0.927 ($P < 0.05$, Fig. 4).

4. Discussion

The dominant bacterium in the normal vaginal microbiota is *Lactobacillus*, which plays a critical role in maintaining vaginal health [9,10]. Owing to environmental conditions, hygiene, and other factors, women are susceptible to reproductive tract infections. If untreated, these infections may progress and ultimately result in severe complications, including pelvic inflammatory disease and endometritis [11]. Currently, vaginal secretions analysis serves as an important diagnostic tool for identifying female reproductive system infections. Traditionally, microscopic examination has been widely used in clinical practice for diagnosis. However, this

Table 7
Analysis of the coincidence rate of detection of *Ureaplasma urealyticum* by qPCR and mNGS diagnostic methods.

Gold standard, (Positive = 55, Negative = 239)	qPCR Detection		mNGS diagnosis	
	Positive	Negative	Positive	Negative
Positive	51	4	52	3
Negative	0	239	0	239
Total	51	243	52	242

Table 8
Value of qPCR and mNGS diagnostic methods in the diagnosis of *Ureaplasma urealyticum*.

Inspection mode	Sensitivity (%)	Specificity (%)	Accuracy (%)	Positive predictive value	Negative predictive value
qPCR Detection	51/51 (100.00)	239/243 (98.35)	290/294 (98.64)	51/55 (92.73)	239/239 (100.00)
mNGS diagnosis	52/52 (100.00)	239/242 (98.76)	291/294 (98.98)	52/55 (94.55)	239/239 (100.00)
χ^2	-	0.000	0.145	0.153	-
P	-	0.996	0.704	0.696	-

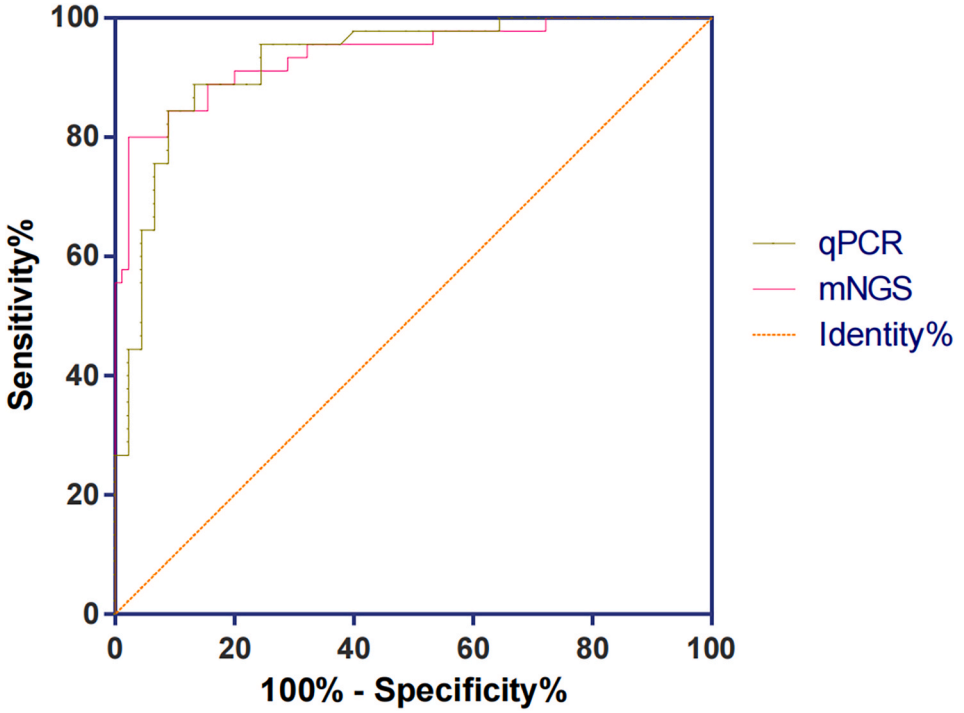


Fig. 4. ROC curve for the diagnosis of *Ureaplasma urealyticum* by qPCR detection and mNGS diagnosis.

method is time-consuming and relies solely on morphological assessment [12,13]. Therefore, there is a clinical need for more efficient and accurate methods, such as molecular detection, to identify pathogens in vaginal secretions.

In this study, we examined vaginal secretions samples from 294 pregnant women, and identified *Staphylococcus aureus*, *Streptococcus agalactiae*, and *Ureaplasma urealyticum* as the three most prevalent pathogens detected in our cohort. Among them, a high detection rate was noted for *S. aureus*, which is often considered conditionally pathogenic or a colonizer in mother-to-neonate transmission [14]. Nevertheless, its presence may signal microbial dysbiosis and it can potentially act synergistically with other pathogens to increase the risk of neonatal infection. Its definitive pathogenic role in high-risk neonates requires confirmation through additional prospective studies. In contrast, *S. agalactiae* is the only Gram-positive coccus conclusively proven to cause invasive neonatal infections [15–17]. *S. agalactiae* is one of the primary causative agents of neonatal sepsis and pneumonia, with its infection potentially leading to severe neonatal disease or even death [18–20]. Furthermore, a relatively high positivity rate was noted for *U. urealyticum,* which may pose a threat to neonatal health. It is noteworthy that although the positivity rates for *Listeria monocytogenes* and *Campylobacter fetus* were relatively low, they were detectable only via qPCR and mNGS. This indicates inherent limitations of conventional bacterial culture methods in detecting certain pathogens (primarily because these pathogens are often difficult to culture clinically or do not grow under standard culture protocols). This is precisely the diagnostic gap that can be bridged using molecular detection technologies [21,22], highlighting their value in integrated diagnostics. Targeted PCR verification experiments based on mNGS data demonstrated that although the sensitivity and accuracy of mNGS for detecting *S. agalactiae* were inferior to those of qPCR, the specificities of both methods were 100%, showing no significant difference from qPCR. ROC curve analysis was used to further validate the efficacy of both assays in diagnosing *S. agalactiae*. Moreover, this study confirmed the sensitivity and specificity of both qPCR and mNGS for detecting *U. urealyticum* and *S. aureus*, demonstrating no significant differences in diagnostic accuracy or predictive value between the two methods. These findings further support the accuracy and reliability of mNGS in detecting vaginal pathogens.

Our results confirm that well-optimized targeted qPCR remains a powerful, cost-effective first-line tool for screening high-risk pregnant women for specific pathogens. Its speed, sensitivity, and low cost are unmatched for this purpose. The complementary

value of mNGS lies in complex scenarios. Its ability to detect *Listeria monocytogenes* and *Campylobacter fetus*—organisms that are difficult to culture routinely—demonstrates its role in closing diagnostic gaps. More broadly, mNGS is indispensable for investigating severe, culture-negative neonatal infections where the pathogen repertoire is unknown or broad, such as in cases of sepsis or meningitis of unknown origin. Here, starting with a maternal sample could provide crucial early clues.

Therefore, we propose a tiered antenatal screening strategy for maternal vaginal pathogenic colonization: 1) Primary Screening: Employ targeted qPCR for core high-risk pathogens (including *Streptococcus agalactiae*, *Staphylococcus aureus*, *Ureaplasma urealyticum*, *Listeria monocytogenes* and *Campylobacter fetus*, which are covered in this study) for routine antenatal screening. This approach is in line with the clinical reality of grassroots and primary medical institutions, with the advantages of low cost, simple operation, short detection cycle, and no requirement for high-end sequencing equipment, which has strong clinical practicability. 2) Reflex Testing with mNGS: Reserve mNGS for high-risk pregnant women (e.g., preterm premature rupture of membranes, clinical chorioamnionitis), culture-negative suspected intrauterine infection cases, or when qPCR results are inconsistent with clinical manifestations, to take full advantage of its unbiased, broad-spectrum detection capability. This strategy follows the clinical diagnostic logic from conventional to complex, from low-cost to high-cost, and can optimize medical resource allocation while ensuring the detection rate of potential pathogens. Notably, the present study excluded pregnant women with PROM, chorioamnionitis and other high-risk conditions to avoid selection bias, and future research will further verify the application value of qPCR and mNGS in these high-risk populations.

It is important to note that this study has some limitations. First, this is a single-center prospective study with a relatively limited sample size, which may constrain the generalizability of the findings to other medical centers or populations. Second, this study only focused on the detection performance of the two molecular methods in maternal vaginal secretions, and did not include paired neonatal clinical outcome data, which means we cannot establish a direct causal relationship between maternal vaginal colonization and neonatal early-onset infection. Third, this study only focused on the detection performance of the two molecular methods in maternal vaginal secretions, and did not include paired neonatal clinical outcome data, which means we cannot establish a direct causal relationship between maternal vaginal colonization and neonatal early-onset infection. Future large-sample, multi-center prospective studies with long-term follow-up of neonatal outcomes (such as sepsis, pneumonia, meningitis, length of hospital stay, and prognosis) are warranted to verify the clinical value of the tiered screening strategy in the prevention and early intervention of early-onset neonatal infection.

5. Conclusion

In summary, this systematic comparison demonstrates that both mNGS and targeted qPCR are highly accurate for detecting common neonatal pathogens in maternal vaginal secretions. While qPCR excels as a routine screening tool, mNGS offers unparalleled breadth for complex cases and rare pathogens. The integration of these technologies into a tiered diagnostic strategy—leveraging the strengths of each—holds significant promise for improving the early diagnosis and management of early-onset neonatal infections. Future work must focus on establishing direct clinical correlations and economic viability to translate this potential into tangible patient benefit.

CRediT authorship contribution statement

Jinyu Zhang: Writing – original draft. **Jingyi Liang:** Data curation, Investigation, Software. **Fei Lv:** Conceptualization, Data curation, Methodology, Project administration. **Zhusheng Guo:** Conceptualization, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare no conflicts of interest.

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Data availability

Data will be made available on request.

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